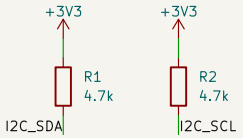
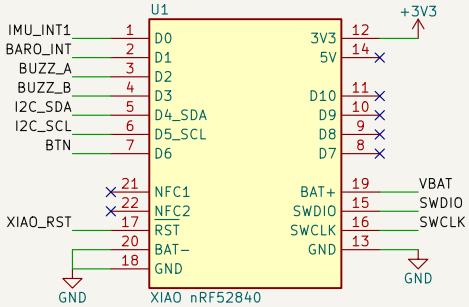
Board notes: keep U2 away from the XIAO antenna end and regulator, under a foam cover with a vent.
Put U3 near the board centre, away from screws. The XIAO antenna end overhangs the carrier or sits over a copper keepout.

I2C pull-ups

To 3V3, i2c1 at 400 kHz.



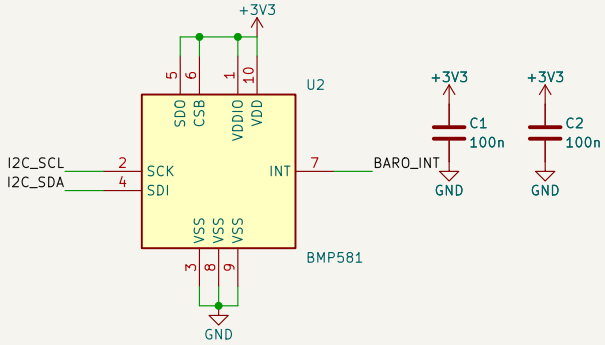
U1 Seeed XIAO nRF52840 (non-Sense)



Onboard: 3.3 V LDO (3V3 pin), BQ25101 LiPo charger, USB-C, UF2 bootloader.
Internal, no carrier connection: P0.31 VBAT_SENSE, P0.14 VBAT_SENSE_EN,
P0.17 CHG_STAT, P0.13 CHG_ISET. Spares: D7-D10 (SPI/UART), NFC1/NFC2.

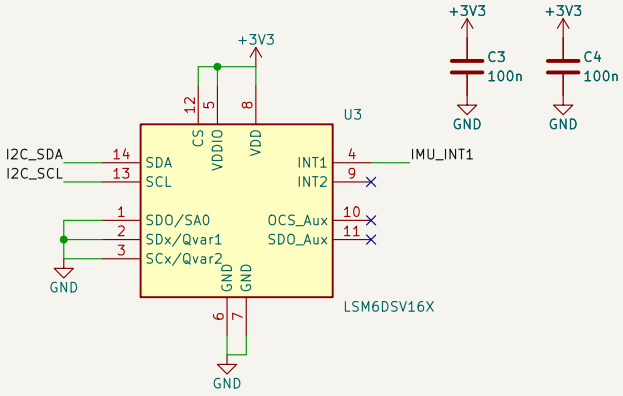
U2 Barometer (BMP581)

I2C address 0x47: SDO and CSB tied to 3V3.
C1 at VDDIO (pin 1), C2 at VDD (pin 10), each close to its pin.



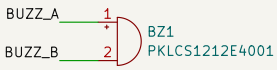
U3 IMU (LSM6DSV16X)

I2C address 0x6A: SDO/SA0 to GND, CS to 3V3.
C3 at VDDIO (pin 5), C4 at VDD (pin 8). Qvar pins 2-3 to GND.
INT2 and aux SPI (pins 10-11) unused.



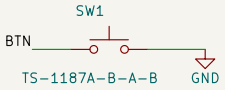
BZ1 Piezo beeper

Driven differentially by PWM1 ch0/ch1, about 6.6 Vpp.



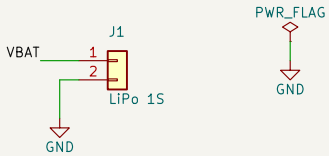
SW1 User / power button

To GND, internal pull-up. Wakes from System OFF.



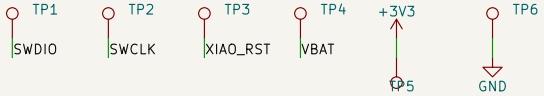
J1 1S LiPo, about 150 mAh

Pin 1 = +. Check the cell lead polarity before plugging in.
Charged by the XIAO through its BAT+ / BAT- underside pads.



Debug pads

SWD for a J-Link or Pico debug probe, wired to the XIAO underside pads.



Pin map: docs/airio-parts-and-pinmap.md
XIAO nRF52840 carrier: BMP581 baro, LSM6DSV16X IMU, piezo, button, LiPo

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Title: airio

Size: A4	Date: 2026-10-24	**Rev: 0.1**
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